

HIL Testbed — Consolidated Diagnostics Report

RPi/HIL directory: D:\My Files\Personal Projects\HIL-Testbed\outputs

(RPi)\runs\52eea3cd-ec2a-48ea-8158-955b51f51d36\publisher

Laptop/dry-run directory: D:\My Files\Personal

Projects\HIL-Testbed\runs\642c8364-50af-4dbd-8934-5c066ff42ddd\publisher

Script	Status
Timing validation + analysis	OK
Outlier detection	OK
Scenario comparison	OK
Q_initial clip diagnostic (H1/H2)	OK
Coordination ground-truth check	OK

Timing validation + analysis

TIMING DATA VALIDATION + ANALYSIS

Loading RPi data from: D:\My Files\Personal Projects\HIL-Testbed\outputs (RPi)\runs\52eea3cd-ec2a-48ea-8158-955b51f51d3
Loading laptop data from: D:\My Files\Personal Projects\HIL-Testbed\runs\642c8364-50af-4dbd-8934-5c066ff42ddd\publisher

VALIDATION CHECKS

16 issue(s) found:

- [RPi/baseline] t_total_ms populated for 35135/35136 timesteps
- [RPi/baseline] 16 values exceed 10x-IQR fence (38.02 ms) – max=2836.01 ms (heuristic flag, not necessarily a bug – inspec)
- [RPi/oltc] 268 values exceed 10x-IQR fence (37.77 ms) – max=1926.41 ms (heuristic flag, not necessarily a bug – inspec)
- [RPi/svc] 5 values exceed 10x-IQR fence (80.15 ms) – max=3307.17 ms (heuristic flag, not necessarily a bug – inspec)
- [RPi/volt_var_local] 2125 exactly-zero t_total_ms values (suspicious)
- [RPi/volt_var_local] 7 values exceed 10x-IQR fence (261.34 ms) – max=5396.60 ms (heuristic flag, not necessarily a bug – inspec)
- [RPi/volt_var_coord] 2125 exactly-zero t_total_ms values (suspicious)
- [RPi/volt_var_coord] 1 values exceed 10x-IQR fence (350.26 ms) – max=7500.84 ms (heuristic flag, not necessarily a bug – inspec)
- [Laptop/baseline] t_total_ms populated for 35135/35136 timesteps
- [Laptop/baseline] 76 values exceed 10x-IQR fence (143.42 ms) – max=5323.83 ms (heuristic flag, not necessarily a bug – inspec)
- [Laptop/oltc] 11 values exceed 10x-IQR fence (151.15 ms) – max=2727.27 ms (heuristic flag, not necessarily a bug – inspec)
- [Laptop/svc] 12 values exceed 10x-IQR fence (301.81 ms) – max=4632.90 ms (heuristic flag, not necessarily a bug – inspec)
- [Laptop/volt_var_local] 2125 exactly-zero t_total_ms values (suspicious)
- [Laptop/volt_var_local] 7 values exceed 10x-IQR fence (346.01 ms) – max=6501.05 ms (heuristic flag, not necessarily a bug – inspec)
- [Laptop/volt_var_coord] 2125 exactly-zero t_total_ms values (suspicious)
- [Laptop/volt_var_coord] 15 values exceed 10x-IQR fence (362.30 ms) – max=9179.62 ms (heuristic flag, not necessarily a bug – inspec)

ANALYSIS

--- A. Per-scenario t_total_ms distribution [RPi] (ms) ---

baseline:

[Full distribution, all timesteps incl. t=0]
n=35135 mean=31.843 std=15.295 min=31.126 max=2836.011
median=31.504 p5=31.318 p25=31.404 p75=32.005 p95=32.310 p99=32.969
First timestep (t=0): 2836.011 ms (90.0x steady-state median)
[Steady-state distribution, excludes first timestep only]
n=35134 mean=31.764 std=3.180 min=31.126 max=201.553
median=31.504 p5=31.318 p25=31.404 p75=32.005 p95=32.310 p99=32.968

oltc:

[Full distribution, all timesteps incl. t=0]
n=35136 mean=34.612 std=18.793 min=33.846 max=1926.412
median=34.238 p5=34.043 p25=34.138 p75=34.468 p95=35.037 p99=37.290
First timestep (t=0): 75.379 ms (2.2x steady-state median)
[Steady-state distribution, excludes first timestep only]
n=35135 mean=34.611 std=18.792 min=33.846 max=1926.412
median=34.238 p5=34.043 p25=34.138 p75=34.468 p95=35.037 p99=37.283

svc:

[Full distribution, all timesteps incl. t=0]
n=35136 mean=66.775 std=34.043 min=65.124 max=3307.170
median=66.038 p5=65.482 p25=65.673 p75=66.989 p95=67.740 p99=69.640
First timestep (t=0): 67.507 ms (1.0x steady-state median)
[Steady-state distribution, excludes first timestep only]
n=35135 mean=66.775 std=34.044 min=65.124 max=3307.170
median=66.038 p5=65.482 p25=65.673 p75=66.989 p95=67.740 p99=69.640

volt_var_local:

[Full distribution, all timesteps incl. t=0]
n=35136 mean=217.006 std=77.886 min=0.000 max=5396.599
median=229.945 p5=0.000 p25=228.080 p75=231.104 p95=233.679 p99=245.239
First timestep (t=0): 231.324 ms (1.0x steady-state median)
[Steady-state distribution, excludes first timestep only]
n=35135 mean=217.005 std=77.887 min=0.000 max=5396.599
median=229.945 p5=0.000 p25=228.080 p75=231.104 p95=233.679 p99=245.239

volt_var_coord:

[Full distribution, all timesteps incl. t=0]
n=35136 mean=232.003 std=72.661 min=0.000 max=7500.843
median=237.972 p5=0.000 p25=235.158 p75=245.622 p95=286.589 p99=298.527
First timestep (t=0): 239.428 ms (1.0x steady-state median)
[Steady-state distribution, excludes first timestep only]
n=35135 mean=232.003 std=72.662 min=0.000 max=7500.843
median=237.972 p5=0.000 p25=235.157 p75=245.622 p95=286.590 p99=298.527

--- A. Per-scenario t_total_ms distribution [Laptop] (ms) ---

baseline:

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[Full distribution, all timesteps incl. t=0]
n=35135 mean=51.361 std=31.017 min=32.179 max=5323.826
median=51.224 p5=37.618 p25=46.423 p75=55.241 p95=60.632 p99=71.501
First timestep (t=0): 5323.826 ms (103.9x steady-state median)
[Steady-state distribution, excludes first timestep only]
n=35134 mean=51.211 std=13.069 min=32.179 max=354.195
median=51.224 p5=37.618 p25=46.423 p75=55.240 p95=60.628 p99=71.485
oltc:
[Full distribution, all timesteps incl. t=0]
n=35136 mean=55.445 std=38.290 min=20.788 max=2727.271
median=55.397 p5=41.653 p25=50.557 p75=59.702 p95=65.025 p99=69.836
First timestep (t=0): 105.753 ms (1.9x steady-state median)
[Steady-state distribution, excludes first timestep only]
n=35135 mean=55.444 std=38.290 min=20.788 max=2727.271
median=55.397 p5=41.653 p25=50.557 p75=59.702 p95=65.024 p99=69.807
svc:
[Full distribution, all timesteps incl. t=0]
n=35136 mean=104.061 std=70.101 min=67.379 max=4632.899
median=104.021 p5=77.883 p25=93.567 p75=112.499 p95=122.351 p99=134.959
First timestep (t=0): 109.845 ms (1.1x steady-state median)
[Steady-state distribution, excludes first timestep only]
n=35135 mean=104.060 std=70.102 min=67.379 max=4632.899
median=104.020 p5=77.883 p25=93.567 p75=112.499 p95=122.351 p99=134.959
volt_var_local:
[Full distribution, all timesteps incl. t=0]
n=35136 mean=108.449 std=77.020 min=0.000 max=6501.048
median=113.149 p5=0.000 p25=101.799 p75=123.999 p95=137.889 p99=150.154
First timestep (t=0): 123.911 ms (1.1x steady-state median)
[Steady-state distribution, excludes first timestep only]
n=35135 mean=108.448 std=77.021 min=0.000 max=6501.048
median=113.149 p5=0.000 p25=101.798 p75=123.999 p95=137.889 p99=150.155
volt_var_coord:
[Full distribution, all timesteps incl. t=0]
n=35136 mean=118.793 std=113.919 min=0.000 max=9179.615
median=123.845 p5=0.000 p25=111.369 p75=134.181 p95=149.321 p99=173.060
First timestep (t=0): 629.046 ms (5.1x steady-state median)
[Steady-state distribution, excludes first timestep only]
n=35135 mean=118.779 std=113.888 min=0.000 max=9179.615
median=123.844 p5=0.000 p25=111.369 p75=134.181 p95=149.316 p99=172.999

--- B. Scenario 4 decomposition: t_total_ms - hil_latency_ms [Rpi] (ms) ---
volt_var_coord (non-serial cost = PF solves + Schur solve + dynamics):
n=35136 mean=87.497 std=48.033 min=0.000 max=7347.144
median=83.681 p5=0.000 p25=81.725 p75=91.697 p95=132.772 p99=144.844
share of t_total_ms that is serial latency: 62.3%
volt_var_local (non-serial cost = PF solves + Schur solve + dynamics):
n=35136 mean=72.530 std=58.069 min=0.000 max=5244.644
median=75.614 p5=0.000 p25=75.224 p75=76.556 p95=79.325 p99=91.280
share of t_total_ms that is serial latency: 66.6%

--- B. Scenario 4 decomposition: t_total_ms - hil_latency_ms [Laptop] (ms) ---
volt_var_coord (non-serial cost = PF solves + Schur solve + dynamics):
n=35136 mean=118.793 std=113.919 min=0.000 max=9179.615
median=123.845 p5=0.000 p25=111.369 p75=134.181 p95=149.321 p99=173.060
share of t_total_ms that is serial latency: 0.0%
volt_var_local (non-serial cost = PF solves + Schur solve + dynamics):
n=35136 mean=108.449 std=77.020 min=0.000 max=6501.048
median=113.149 p5=0.000 p25=101.799 p75=123.999 p95=137.889 p99=150.154
share of t_total_ms that is serial latency: 0.0%

--- C. HIL vs dry-run t_total_ms comparison (ms) ---
baseline:
Rpi : mean=31.843 ms median=31.504 ms
Laptop : mean=51.361 ms median=51.224 ms
Rpi/Laptop ratio (mean): 0.62x
oltc:
Rpi : mean=34.612 ms median=34.238 ms
Laptop : mean=55.445 ms median=55.397 ms
Rpi/Laptop ratio (mean): 0.62x
svc:
Rpi : mean=66.775 ms median=66.038 ms
Laptop : mean=104.061 ms median=104.021 ms
Rpi/Laptop ratio (mean): 0.64x
volt_var_local:
Rpi : mean=217.006 ms median=229.945 ms
Laptop : mean=108.449 ms median=113.149 ms
Rpi/Laptop ratio (mean): 2.00x
volt_var_coord:
Rpi : mean=232.003 ms median=237.972 ms
Laptop : mean=118.793 ms median=123.845 ms
Rpi/Laptop ratio (mean): 1.95x

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--- D. Correlation with control activity [RPI] ---

OLTC – tap_changed vs tap_unchanged t_total_ms:

tap_changed=True : n=1 mean=75.379 ms
tap_changed=False : n=35135 mean=34.611 ms

SVC – svc_saturated vs not t_total_ms:

saturated=True : n=35127 mean=66.775 ms
saturated=False : n=9 mean=67.150 ms

volt_var_coord – coordination_active / curtailment_needed:

coordination_active=True : n=33008 mean=246.937 ms
coordination_active=False : n=2128 mean=0.353 ms
curtailment_needed=True : n=16729 mean=248.877 ms
curtailment_needed=False : n=18407 mean=216.668 ms

volt_var_local – coordination_active / curtailment_needed:

coordination_active=True : n=0
coordination_active=False : n=35136 mean=217.006 ms
curtailment_needed=True : n=16729 mean=230.870 ms
curtailment_needed=False : n=18407 mean=204.405 ms

--- D. Correlation with control activity [Laptop] ---

OLTC – tap_changed vs tap_unchanged t_total_ms:

tap_changed=True : n=1 mean=105.753 ms
tap_changed=False : n=35135 mean=55.444 ms

SVC – svc_saturated vs not t_total_ms:

saturated=True : n=35127 mean=104.062 ms
saturated=False : n=9 mean=98.064 ms

volt_var_coord – coordination_active / curtailment_needed:

coordination_active=True : n=2277 mean=126.675 ms
coordination_active=False : n=32859 mean=118.247 ms
curtailment_needed=True : n=16727 mean=127.486 ms
curtailment_needed=False : n=18409 mean=110.894 ms

volt_var_local – coordination_active / curtailment_needed:

coordination_active=True : n=0
coordination_active=False : n=35136 mean=108.449 ms
curtailment_needed=True : n=16727 mean=114.518 ms
curtailment_needed=False : n=18409 mean=102.934 ms

--- E. Outlier scan + temporal trend check [RPI] ---

baseline:

Top 5 slowest timesteps:

t=0 t_total_ms=2836.011, converged=True, curtailment_needed=False
t=1704 t_total_ms=201.553, converged=True, curtailment_needed=False
t=15829 t_total_ms=188.281, converged=True, curtailment_needed=False
t=6391 t_total_ms=186.930, converged=True, curtailment_needed=False
t=8764 t_total_ms=184.387, converged=True, curtailment_needed=False

Temporal trend: slope=-0.000004 ms/timestep, corr(t, t_total_ms)=-0.0028

Implied drift over this run: -0.149 ms (-0.5% relative to mean)

No strong linear trend detected ($|corr| \leq 0.3$) – variation looks more like noise/isolated spikes than systematic

Quartile-block means (chronological): Q1=31.885, Q2=31.707, Q3=32.084, Q4=31.698

oltc:

Top 5 slowest timesteps:

t=29114 t_total_ms=1926.412, converged=True, tap_changed=False, tap_attempted=False, curtailment_needed=False
t=21564 t_total_ms=1735.977, converged=True, tap_changed=False, tap_attempted=False, curtailment_needed=False
t=14618 t_total_ms=1570.179, converged=True, tap_changed=False, tap_attempted=False, curtailment_needed=False
t=8275 t_total_ms=1411.864, converged=True, tap_changed=False, tap_attempted=False, curtailment_needed=False
t=2333 t_total_ms=1326.281, converged=True, tap_changed=False, tap_attempted=False, curtailment_needed=False

Temporal trend: slope=-0.000000 ms/timestep, corr(t, t_total_ms)=-0.0002

Implied drift over this run: -0.015 ms (-0.0% relative to mean)

No strong linear trend detected ($|corr| \leq 0.3$) – variation looks more like noise/isolated spikes than systematic

Quartile-block means (chronological): Q1=34.606, Q2=34.637, Q3=34.604, Q4=34.601

svc:

Top 5 slowest timesteps:

t=28628 t_total_ms=3307.170, converged=True, svc_saturated=True, curtailment_needed=False
t=20927 t_total_ms=3072.325, converged=True, svc_saturated=True, curtailment_needed=False
t=13898 t_total_ms=2827.102, converged=True, svc_saturated=True, curtailment_needed=False
t=1487 t_total_ms=2723.542, converged=True, svc_saturated=True, curtailment_needed=False
t=7424 t_total_ms=2611.165, converged=True, svc_saturated=True, curtailment_needed=False

Temporal trend: slope=-0.000046 ms/timestep, corr(t, t_total_ms)=-0.0138

Implied drift over this run: -1.631 ms (-2.4% relative to mean)

No strong linear trend detected ($|corr| \leq 0.3$) – variation looks more like noise/isolated spikes than systematic

Quartile-block means (chronological): Q1=67.836, Q2=66.457, Q3=66.269, Q4=66.537

volt_var_local:

Top 5 slowest timesteps:

t=30640 t_total_ms=5396.599, converged=True, coordination_active=False, curtailment_needed=True
t=24138 t_total_ms=5107.657, converged=True, coordination_active=False, curtailment_needed=True
t=17959 t_total_ms=4815.674, converged=True, coordination_active=False, curtailment_needed=True
t=12023 t_total_ms=4552.309, converged=True, coordination_active=False, curtailment_needed=True
t=837 t_total_ms=4243.417, converged=True, coordination_active=False, curtailment_needed=True

Temporal trend: slope=0.000586 ms/timestep, corr(t, t_total_ms)=0.0763

Implied drift over this run: +20.598 ms (+0.5% relative to mean)

No strong linear trend detected ($|corr| \leq 0.3$) – variation looks more like noise/isolated spikes than systematic

Quartile-block means (chronological): Q1=195.686, Q2=229.859, Q3=229.580, Q4=212.898

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volt_var_coord:
  Top 5 slowest timesteps:
    t=26386 t_total_ms=7500.843, converged=True, coordination_active=True, curtailment_needed=False
    t=24854 t_total_ms=326.176, converged=True, coordination_active=True, curtailment_needed=True
    t=25138 t_total_ms=316.858, converged=True, coordination_active=True, curtailment_needed=True
    t=34755 t_total_ms=316.478, converged=True, coordination_active=True, curtailment_needed=True
    t=24706 t_total_ms=316.282, converged=True, coordination_active=True, curtailment_needed=False
  Temporal trend: slope=0.001594 ms/timestep, corr(t, t_total_ms)=0.2225
  Implied drift over this run: +56.007 ms (+24.1% relative to mean)
  No strong linear trend detected (|corr| <= 0.3) – variation looks more like noise/isolated spikes than systematic
  Quartile-block means (chronological): Q1=201.979, Q2=237.114, Q3=241.349, Q4=247.571

--- E. Outlier scan + temporal trend check [Laptop] ---
baseline:
  Top 5 slowest timesteps:
    t=0 t_total_ms=5323.826, converged=True, curtailment_needed=False
    t=819 t_total_ms=354.195, converged=True, curtailment_needed=False
    t=11984 t_total_ms=339.904, converged=True, curtailment_needed=False
    t=4978 t_total_ms=329.310, converged=True, curtailment_needed=False
    t=2385 t_total_ms=312.568, converged=True, curtailment_needed=False
  Temporal trend: slope=-0.000090 ms/timestep, corr(t, t_total_ms)=-0.0294
  Implied drift over this run: -3.154 ms (-6.1% relative to mean)
  No strong linear trend detected (|corr| <= 0.3) – variation looks more like noise/isolated spikes than systematic
  Quartile-block means (chronological): Q1=53.101, Q2=50.846, Q3=50.407, Q4=51.089
oltc:
  Top 5 slowest timesteps:
    t=31908 t_total_ms=2727.271, converged=True, tap_changed=False, tap_attempted=False, curtailment_needed=False
    t=27428 t_total_ms=2721.822, converged=True, tap_changed=False, tap_attempted=False, curtailment_needed=False
    t=23202 t_total_ms=2572.224, converged=True, tap_changed=False, tap_attempted=False, curtailment_needed=False
    t=19205 t_total_ms=2528.882, converged=True, tap_changed=False, tap_attempted=False, curtailment_needed=False
    t=15413 t_total_ms=2422.301, converged=True, tap_changed=False, tap_attempted=False, curtailment_needed=False
  Temporal trend: slope=-0.000133 ms/timestep, corr(t, t_total_ms)=-0.0353
  Implied drift over this run: -4.685 ms (-8.4% relative to mean)
  No strong linear trend detected (|corr| <= 0.3) – variation looks more like noise/isolated spikes than systematic
  Quartile-block means (chronological): Q1=56.841, Q2=55.885, Q3=55.530, Q4=53.525
svc:
  Top 5 slowest timesteps:
    t=33127 t_total_ms=4632.899, converged=True, svc_saturated=True, curtailment_needed=False
    t=29287 t_total_ms=4547.518, converged=True, svc_saturated=True, curtailment_needed=False
    t=25597 t_total_ms=4303.026, converged=True, svc_saturated=True, curtailment_needed=False
    t=18633 t_total_ms=4226.152, converged=True, svc_saturated=True, curtailment_needed=False
    t=22046 t_total_ms=4121.904, converged=True, svc_saturated=True, curtailment_needed=False
  Temporal trend: slope=0.000140 ms/timestep, corr(t, t_total_ms)=0.0202
  Implied drift over this run: +4.909 ms (+4.7% relative to mean)
  No strong linear trend detected (|corr| <= 0.3) – variation looks more like noise/isolated spikes than systematic
  Quartile-block means (chronological): Q1=104.927, Q2=100.019, Q3=105.570, Q4=105.727
volt_var_local:
  Top 5 slowest timesteps:
    t=33187 t_total_ms=6501.048, converged=True, coordination_active=False, curtailment_needed=True
    t=24062 t_total_ms=6309.090, converged=True, coordination_active=False, curtailment_needed=False
    t=26061 t_total_ms=6109.389, converged=True, coordination_active=False, curtailment_needed=False
    t=14964 t_total_ms=5807.216, converged=True, coordination_active=False, curtailment_needed=False
    t=3129 t_total_ms=5263.548, converged=True, coordination_active=False, curtailment_needed=False
  Temporal trend: slope=0.000012 ms/timestep, corr(t, t_total_ms)=0.0016
  Implied drift over this run: +0.436 ms (+0.4% relative to mean)
  No strong linear trend detected (|corr| <= 0.3) – variation looks more like noise/isolated spikes than systematic
  Quartile-block means (chronological): Q1=100.952, Q2=116.053, Q3=117.239, Q4=99.552
volt_var_coord:
  Top 5 slowest timesteps:
    t=23244 t_total_ms=9179.615, converged=True, coordination_active=False, curtailment_needed=True
    t=25941 t_total_ms=9061.134, converged=True, coordination_active=False, curtailment_needed=False
    t=17210 t_total_ms=8223.983, converged=True, coordination_active=False, curtailment_needed=False
    t=11455 t_total_ms=7954.115, converged=True, coordination_active=False, curtailment_needed=False
    t=2350 t_total_ms=7929.738, converged=True, coordination_active=False, curtailment_needed=False
  Temporal trend: slope=0.000342 ms/timestep, corr(t, t_total_ms)=0.0305
  Implied drift over this run: +12.030 ms (+10.1% relative to mean)
  No strong linear trend detected (|corr| <= 0.3) – variation looks more like noise/isolated spikes than systematic
  Quartile-block means (chronological): Q1=109.993, Q2=124.155, Q3=118.229, Q4=122.795

--- F. Run coverage [RPI] ---
  This run covers 35136 timesteps (full year (366 days), 15-min resolution). All seasons are represented. Distribution

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END OF REPORT
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Outlier detection

OUTLIER DIAGNOSTIC FOLLOW-UP

Loading RPi data from: D:\My Files\Personal Projects\HIL-Testbed\outputs (RPi)\runs\52eea3cd-ec2a-48ea-8158-955b51f51d3
Loading laptop data from: D:\My Files\Personal Projects\HIL-Testbed\runs\642c8364-50af-4dbd-8934-5c066ff42ddd\publisher

ENVIRONMENT: RPi

--- baseline: 15 outlier(s) (t=0 excluded) ---

Top 15 outliers:

t=1704, t_total_ms=201.5532330115093, timestamp=2016-01-18T18:00:00+01:00, converged=True, tap_changed=None, tap_at=1704, t_total_ms=188.28116600343492, timestamp=2016-06-13T22:15:00+02:00, converged=True, tap_changed=None, tap_at=15829, t_total_ms=186.92998299957253, timestamp=2016-03-07T13:45:00+01:00, converged=True, tap_changed=None, tap_at=6391, t_total_ms=184.38728799810633, timestamp=2016-04-01T08:00:00+02:00, converged=True, tap_changed=None, tap_at=8764, t_total_ms=184.08956499479245, timestamp=2016-07-08T03:15:00+02:00, converged=True, tap_changed=None, tap_at=18153, t_total_ms=183.4797830088064, timestamp=2016-09-18T20:15:00+02:00, converged=True, tap_changed=None, tap_at=25133, t_total_ms=183.19925099785905, timestamp=2016-11-29T21:00:00+01:00, converged=True, tap_changed=None, tap_at=32052, t_total_ms=182.61432800500188, timestamp=2016-12-24T02:00:00+01:00, converged=True, tap_changed=None, tap_at=34376, t_total_ms=182.0086759980768, timestamp=2016-08-01T17:30:00+02:00, converged=True, tap_changed=None, tap_at=20514, t_total_ms=181.9341710070148, timestamp=2016-04-26T01:00:00+02:00, converged=True, tap_changed=None, tap_at=11136, t_total_ms=181.8323970073834, timestamp=2016-10-12T19:30:00+02:00, converged=True, tap_changed=None, tap_at=27434, t_total_ms=181.2325229984708, timestamp=2016-08-26T11:30:00+02:00, converged=True, tap_changed=None, tap_at=22890, t_total_ms=180.77242000435945, timestamp=2016-05-19T23:00:00+02:00, converged=True, tap_changed=None, tap_at=13432, t_total_ms=180.27463000908028, timestamp=2016-02-12T10:30:00+01:00, converged=True, tap_changed=None, tap_at=4074, t_total_ms=179.80786599218845, timestamp=2016-11-06T18:15:00+01:00, converged=True, tap_changed=None, tap_at=29833

[RPi/baseline] Periodicity check:

n=15 outliers, t range [1704, 34376]

Gaps between consecutive outlier timesteps: mean=2333.7 std=53.1 min=2219 max=2399

Coefficient of variation of gaps: 0.023

OBSERVATION: gaps are highly regular (CV < 0.15) – outliers occur at near-evenly-spaced intervals. This pattern is

[RPi/baseline] Calendar placement of outliers:

Outlier count by month:

Month 1: 1
Month 2: 1
Month 3: 1
Month 4: 2
Month 5: 1
Month 6: 1
Month 7: 1
Month 8: 2
Month 9: 1
Month 10: 1
Month 11: 2
Month 12: 1

--- oltc: 267 outlier(s) (t=0 excluded) ---

Top 20 outliers:

t=29114, t_total_ms=1926.4115620026132, timestamp=2016-10-30T06:30:00+01:00, converged=True, tap_changed=False, tap_at=29114, t_total_ms=1735.9773859934648, timestamp=2016-08-12T16:00:00+02:00, converged=True, tap_changed=False, tap_at=21564, t_total_ms=1570.1790739985881, timestamp=2016-06-01T07:30:00+02:00, converged=True, tap_changed=False, tap_at=14618, t_total_ms=1411.8637070059776, timestamp=2016-03-27T05:45:00+02:00, converged=True, tap_changed=False, tap_at=8275, t_total_ms=1326.2813210021704, timestamp=2016-01-25T07:15:00+01:00, converged=True, tap_changed=False, tap_at=2333, t_total_ms=41.318236006191, timestamp=2016-07-31T22:15:00+02:00, converged=True, tap_changed=False, tap_at=20437, t_total_ms=41.186488000676036, timestamp=2016-05-29T16:30:00+02:00, converged=True, tap_changed=False, tap_at=14366, t_total_ms=41.101279988652095, timestamp=2016-08-16T09:30:00+02:00, converged=True, tap_changed=False, tap_at=21922, t_total_ms=40.78317301173229, timestamp=2016-08-22T14:45:00+02:00, converged=True, tap_changed=False, tap_at=22519, t_total_ms=40.44353299832437, timestamp=2016-04-01T08:00:00+02:00, converged=True, tap_changed=False, tap_at=8764, t_total_ms=40.418183998554014, timestamp=2016-06-07T09:15:00+02:00, converged=True, tap_changed=False, tap_at=15201, t_total_ms=40.26618500938639, timestamp=2016-05-03T02:00:00+02:00, converged=True, tap_changed=False, tap_at=11812, t_total_ms=40.09948900784366, timestamp=2016-08-12T13:45:00+02:00, converged=True, tap_changed=False, tap_at=21555, t_total_ms=40.082118997815996, timestamp=2016-08-13T20:45:00+02:00, converged=True, tap_changed=False, tap_at=21679, t_total_ms=39.78822100907564, timestamp=2016-12-21T08:00:00+01:00, converged=True, tap_changed=False, tap_at=34112, t_total_ms=39.520207996247336, timestamp=2016-06-02T08:30:00+02:00, converged=True, tap_changed=False, tap_at=14718, t_total_ms=39.50300099677406, timestamp=2016-09-08T01:30:00+02:00, converged=True, tap_changed=False, tap_at=24098, t_total_ms=39.48346301331185, timestamp=2016-05-25T16:15:00+02:00, converged=True, tap_changed=False, tap_at=13981, t_total_ms=39.481460000388324, timestamp=2016-04-16T12:15:00+02:00, converged=True, tap_changed=False, tap_at=10221, t_total_ms=39.43071101093665, timestamp=2016-07-25T13:45:00+02:00, converged=True, tap_changed=False, tap_at=19827

[RPi/oltc] Periodicity check:

n=267 outliers, t range [97, 35081]

Gaps between consecutive outlier timesteps: mean=131.5 std=57.9 min=1 max=495

Coefficient of variation of gaps: 0.440

OBSERVATION: gaps are irregular (CV >= 0.4) – no strong evidence of periodicity; outliers look more scattered/random

```

[RPi/oltc] Root-cause signature check:
  Of 267 outlier timesteps: 0 had tap_attempted=True, 267 had tap_attempted=False
  OBSERVATION: majority of slow timesteps did NOT involve a tap attempt – the slowdown is occurring in the base power

[RPi/oltc] Calendar placement of outliers:
  Outlier count by month:
    Month 1: 16
    Month 2: 17
    Month 3: 21
    Month 4: 26
    Month 5: 24
    Month 6: 26
    Month 7: 21
    Month 8: 21
    Month 9: 23
    Month 10: 25
    Month 11: 21
    Month 12: 26

--- svc: 5 outlier(s) (t=0 excluded) ---
Top 5 outliers:
  t=28628, t_total_ms=3307.1701860026224, timestamp=2016-10-25T06:00:00+02:00, converged=True, tap_changed=None, tap_a
  t=20927, t_total_ms=3072.325169996475, timestamp=2016-08-06T00:45:00+02:00, converged=True, tap_changed=None, tap_a
  t=13898, t_total_ms=2827.10244700138, timestamp=2016-05-24T19:30:00+02:00, converged=True, tap_changed=None, tap_at
  t=1487, t_total_ms=2723.5420890065143, timestamp=2016-01-16T11:45:00+01:00, converged=True, tap_changed=None, tap_a
  t=7424, t_total_ms=2611.164754998754, timestamp=2016-03-18T08:00:00+01:00, converged=True, tap_changed=None, tap_at

[RPi/svc] Periodicity check:
  n=5 outliers, t range [1487, 28628]
  Gaps between consecutive outlier timesteps: mean=6785.2 std=654.7 min=5937 max=7701
  Coefficient of variation of gaps: 0.096
  OBSERVATION: gaps are highly regular (CV < 0.15) – outliers occur at near-evenly-spaced intervals. This pattern is

[RPi/svc] Calendar placement of outliers:
  Outlier count by month:
    Month 1: 1
    Month 3: 1
    Month 5: 1
    Month 8: 1
    Month 10: 1

--- volt_var_local: 7 outlier(s) (t=0 excluded) ---
Top 7 outliers:
  t=30640, t_total_ms=5396.599201994832, timestamp=2016-11-15T04:00:00+01:00, converged=True, tap_changed=None, tap_a
  t=24138, t_total_ms=5107.657343993196, timestamp=2016-09-08T11:30:00+02:00, converged=True, tap_changed=None, tap_a
  t=17959, t_total_ms=4815.674347002641, timestamp=2016-07-06T02:45:00+02:00, converged=True, tap_changed=None, tap_a
  t=12023, t_total_ms=4552.308767000795, timestamp=2016-05-05T06:45:00+02:00, converged=True, tap_changed=None, tap_a
  t=837, t_total_ms=4243.416957004229, timestamp=2016-01-09T17:15:00+01:00, converged=True, tap_changed=None, tap_att
  t=31225, t_total_ms=264.41148000594694, timestamp=2016-11-21T06:15:00+01:00, converged=True, tap_changed=None, tap_a
  t=29026, t_total_ms=263.76990700373426, timestamp=2016-10-29T09:30:00+02:00, converged=True, tap_changed=None, tap_a

[RPi/volt_var_local] Periodicity check:
  n=7 outliers, t range [837, 31225]
  Gaps between consecutive outlier timesteps: mean=5064.7 std=3451.5 min=585 max=11186
  Coefficient of variation of gaps: 0.681
  OBSERVATION: gaps are irregular (CV >= 0.4) – no strong evidence of periodicity; outliers look more scattered/random

[RPi/volt_var_local] Calendar placement of outliers:
  Outlier count by month:
    Month 1: 1
    Month 5: 1
    Month 7: 1
    Month 9: 1
    Month 10: 1
    Month 11: 2

--- volt_var_coord: 1 outlier(s) (t=0 excluded) ---
Top 1 outliers:
  t=26386, t_total_ms=7500.842605004436, timestamp=2016-10-01T21:30:00+02:00, converged=True, tap_changed=None, tap_a

[RPi/volt_var_coord] Periodicity check:
  Only 1 outlier(s) – too few to assess periodicity.

[RPi/volt_var_coord] Calendar placement of outliers:
  Outlier count by month:
    Month 10: 1
  OBSERVATION: month 10 alone accounts for 100% of outliers – possible seasonal concentration.

[RPi] Cross-scenario outlier clustering (within +/-5 timesteps):
  baseline vs oltc: 3/267 of oltc's outliers (1.1%) fall within 5 steps of an baseline outlier
  Example matches: t=8764~8764, t=22886~22890, t=29831~29833

```

baseline vs svc: 0/5 of svc's outliers (0.0%) fall within 5 steps of an baseline outlier
baseline vs volt_var_local: 0/7 of volt_var_local's outliers (0.0%) fall within 5 steps of an baseline outlier
baseline vs volt_var_coord: 0/1 of volt_var_coord's outliers (0.0%) fall within 5 steps of an baseline outlier
oltc vs svc: 0/5 of svc's outliers (0.0%) fall within 5 steps of an oltc outlier
oltc vs volt_var_local: 1/7 of volt_var_local's outliers (14.3%) fall within 5 steps of an oltc outlier
oltc vs volt_var_coord: 0/1 of volt_var_coord's outliers (0.0%) fall within 5 steps of an oltc outlier
svc vs volt_var_local: 0/7 of volt_var_local's outliers (0.0%) fall within 5 steps of an svc outlier
svc vs volt_var_coord: 0/1 of volt_var_coord's outliers (0.0%) fall within 5 steps of an svc outlier
volt_var_local vs volt_var_coord: 0/1 of volt_var_coord's outliers (0.0%) fall within 5 steps of an volt_var_local

[RPI] Exact-zero t_total_ms anomaly check:

volt_var_coord: 2125 record(s) with t_total_ms == 0.0

t=138, full record:

t = 138

timestamp = 2016-01-02T10:30:00+01:00

converged = True

max_vm_pu = 1.058506022659314

min_vm_pu = 1.0185948343717153

max_line_loading_pct = 64.4036247035777

max_trafo_loading_pct = 25.813793991747406

line_loading_pct = {'0': 40.537806494740195, '1': 38.80370748712361, '2': 37.07174586398684, '3': 14.1706635106}

trafo_loading_pct = {'0': 25.813793991747406, '1': 25.813793991747406}

over_voltage_buses = []

under_voltage_buses = []

overloaded_lines = []

overloaded_trafos = []

violation_flag = False

tap_pos = None

tap_changed = None

tap_attempted = None

tap_blocked_reason = None

svc_q_mvar = None

svc_saturated = None

curtailment_needed = False

curtail_exhausted = False

coordination_active = False

q_saturated_count = 0

hil_latency_ms = 0.0

t_total_ms = 0.0

losses_mw = 0.2847817712814325

grid_import_mw = -12.79875113943856

der_gen_mw = 7.73652226672

load_mw = 12.085489356

Neighbouring t_total_ms values (t-2..t+2):

t=136: t_total_ms=233.94907200417947

t=137: t_total_ms=234.05423799704295

t=138: t_total_ms=0.0

t=139: t_total_ms=230.65024300012738

t=140: t_total_ms=233.8542020006571

t=150, full record:

t = 150

timestamp = 2016-01-02T13:30:00+01:00

converged = True

max_vm_pu = 1.0571070010690617

min_vm_pu = 1.0184558127783514

max_line_loading_pct = 54.86247904533704

max_trafo_loading_pct = 22.606504706612867

line_loading_pct = {'0': 33.97599001765698, '1': 31.59701484087789, '2': 29.217729658819113, '3': 11.1718483607}

trafo_loading_pct = {'0': 22.606504706612867, '1': 22.606504706612867}

over_voltage_buses = []

under_voltage_buses = []

overloaded_lines = []

overloaded_trafos = []

violation_flag = False

tap_pos = None

tap_changed = None

tap_attempted = None

tap_blocked_reason = None

svc_q_mvar = None

svc_saturated = None

curtailment_needed = False

curtail_exhausted = False

coordination_active = False

q_saturated_count = 0

hil_latency_ms = 0.0

t_total_ms = 0.0

losses_mw = 0.24306025789650254

grid_import_mw = -11.174554256583509

der_gen_mw = 5.6242198643800005

load_mw = 11.6391053499

Neighbouring t_total_ms values (t-2..t+2):


```

t=148: t_total_ms=237.91382199851796
t=149: t_total_ms=233.98485100187827
t=150: t_total_ms=0.0
t=151: t_total_ms=0.0
t=152: t_total_ms=0.0
t=151, full record:
t = 151
timestamp = 2016-01-02T13:45:00+01:00
converged = True
max_vm_pu = 1.0552353846159048
min_vm_pu = 1.018711928195175
max_line_loading_pct = 52.01563544024469
max_trafo_loading_pct = 22.16044216924768
line_loading_pct = {'0': 32.89992748604985, '1': 30.55772190526117, '2': 28.21522394723013, '3': 10.85416516983}
trafo_loading_pct = {'0': 22.16044216924768, '1': 22.16044216924768}
over_voltage_buses = []
under_voltage_buses = []
overloaded_lines = []
overloaded_trafos = []
violation_flag = False
tap_pos = None
tap_changed = None
tap_attempted = None
tap_blocked_reason = None
svc_q_mvar = None
svc_saturated = None
curtailment_needed = False
curtail_exhausted = False
coordination_active = False
q_saturated_count = 0
hil_latency_ms = 0.0
t_total_ms = 0.0
losses_mw = 0.22712921082239906
grid_import_mw = -10.951948755887575
der_gen_mw = 5.2999206048100005
load_mw = 11.5533426381
Neighbouring t_total_ms values (t-2..t+2):
t=149: t_total_ms=233.98485100187827
t=150: t_total_ms=0.0
t=151: t_total_ms=0.0
t=152: t_total_ms=0.0
t=153: t_total_ms=236.37101899657864
t=152, full record:
t = 152
timestamp = 2016-01-02T14:00:00+01:00
converged = True
max_vm_pu = 1.0567733808493154
min_vm_pu = 1.0207885202584268
max_line_loading_pct = 51.48620659616792
max_trafo_loading_pct = 24.03980660243129
line_loading_pct = {'0': 33.7874802928522, '1': 31.412997765433342, '2': 29.03929061963662, '3': 11.33786167835}
trafo_loading_pct = {'0': 24.03980660243129, '1': 24.03980660243129}
over_voltage_buses = []
under_voltage_buses = []
overloaded_lines = []
overloaded_trafos = []
violation_flag = False
tap_pos = None
tap_changed = None
tap_attempted = None
tap_blocked_reason = None
svc_q_mvar = None
svc_saturated = None
curtailment_needed = False
curtail_exhausted = False
coordination_active = False
q_saturated_count = 0
hil_latency_ms = 0.0
t_total_ms = 0.0
losses_mw = 0.23405673087333956
grid_import_mw = -12.060420718566627
der_gen_mw = 4.97562133809
load_mw = 10.11364388865
Neighbouring t_total_ms values (t-2..t+2):
t=150: t_total_ms=0.0
t=151: t_total_ms=0.0
t=152: t_total_ms=0.0
t=153: t_total_ms=236.37101899657864
t=154: t_total_ms=237.1030500071356
t=158, full record:
t = 158

```

```

timestamp = 2016-01-02T15:30:00+01:00
converged = True
max_vm_pu = 1.058130770304654
min_vm_pu = 1.0211437773171492
max_line_loading_pct = 46.75298446935188
max_trafo_loading_pct = 18.197084937433527
line_loading_pct = {'0': 24.210894444236093, '1': 22.134593704886495, '2': 20.05772570556373, '3': 7.8488488677}
trafo_loading_pct = {'0': 18.197084937433527, '1': 18.197084937433527}
over_voltage_buses = []
under_voltage_buses = []
overloaded_lines = []
overloaded_trafos = []
violation_flag = False
tap_pos = None
tap_changed = None
tap_attempted = None
tap_blocked_reason = None
svc_q_mvar = None
svc_saturated = None
curtailment_needed = False
curtail_exhausted = False
coordination_active = False
q_saturated_count = 0
hil_latency_ms = 0.0
t_total_ms = 0.0
losses_mw = 0.19595177204758793
grid_import_mw = -9.108411212642434
der_gen_mw = 1.1086271654900002
load_mw = 9.236764180800002
Neighbouring t_total_ms values (t-2..t+2):
t=156: t_total_ms=249.07027899462264
t=157: t_total_ms=245.77343299461063
t=158: t_total_ms=0.0
t=159: t_total_ms=0.0
t=160: t_total_ms=0.0
t=159, full record:
t = 159
timestamp = 2016-01-02T15:45:00+01:00
converged = True
max_vm_pu = 1.0558777367610186
min_vm_pu = 1.0206791121921028
max_line_loading_pct = 44.66516745758676
max_trafo_loading_pct = 17.491033282555453
line_loading_pct = {'0': 23.7242306569724, '1': 21.63138618829142, '2': 19.537994107351178, '3': 7.625576908528}
trafo_loading_pct = {'0': 17.491033282555453, '1': 17.491033282555453}
over_voltage_buses = []
under_voltage_buses = []
overloaded_lines = []
overloaded_trafos = []
violation_flag = False
tap_pos = None
tap_changed = None
tap_attempted = None
tap_blocked_reason = None
svc_q_mvar = None
svc_saturated = None
curtailment_needed = False
curtail_exhausted = False
coordination_active = False
q_saturated_count = 0
hil_latency_ms = 0.0
t_total_ms = 0.0
losses_mw = 0.1840745188444281
grid_import_mw = -8.713413224385592
der_gen_mw = 1.03454373828
load_mw = 9.569555995049999
Neighbouring t_total_ms values (t-2..t+2):
t=157: t_total_ms=245.77343299461063
t=158: t_total_ms=0.0
t=159: t_total_ms=0.0
t=160: t_total_ms=0.0
t=161: t_total_ms=0.0
t=160, full record:
t = 160
timestamp = 2016-01-02T16:00:00+01:00
converged = True
max_vm_pu = 1.0550953684838602
min_vm_pu = 1.0199716963351726
max_line_loading_pct = 44.385285099157485
max_trafo_loading_pct = 16.279622881051694
line_loading_pct = {'0': 21.684084553571385, '1': 19.782015533774157, '2': 17.87843785667433, '3': 6.9860843360}

```

Scenario comparison

```
#####
# ENVIRONMENT: RPi - D:\My Files\Personal Projects\HIL-Testbed\outputs (RPi)\runs\52eea3cd-ec2a-48ea-8158-955b51f51d3
#####
```

C. Extreme State Timeseries [RPi]

--- C1. max_vm_pu distribution across scenarios ---

Scenario	mean	median	p95	p99	max	> 1.05 (%)	> 1.08 (%)
Baseline	1.0769	1.0766	1.0962	1.1038	1.1166	99.3%	37.1%
OLTC	1.0622	1.0619	1.0816	1.0893	1.1022	86.5%	6.7%
SVC	1.0278	1.0250	1.0398	1.0486	1.0567	0.6%	0.0%
Volt-Var Local (4A)	1.0357	1.0353	1.0515	1.0591	1.0600	5.3%	0.0%
Volt-Var Coord (4B)	1.0354	1.0352	1.0515	1.0591	1.0600	5.3%	0.0%

--- C2. max_line_loading_pct distribution across scenarios ---

Scenario	mean	median	p95	p99	max	> 100% (steps)
Baseline	78.82	73.15	127.20	141.17	159.74	7515
OLTC	79.93	74.19	128.99	143.14	161.95	7919
SVC	131.31	126.72	161.37	173.82	189.80	35136
Volt-Var Local (4A)	79.78	85.41	89.62	89.92	90.00	0
Volt-Var Coord (4B)	79.77	85.39	89.62	89.92	90.00	0

--- C3. Voltage envelope: how often does max_vm_pu exceed thresholds? ---

Scenario	>= 1.02	>= 1.05	>= 1.08	>= 1.10
Baseline	100.0%	99.3%	37.1%	2.2%
OLTC	100.0%	86.5%	6.7%	0.0%
SVC	100.0%	0.6%	0.0%	0.0%
Volt-Var Local (4A)	100.0%	5.3%	0.0%	0.0%
Volt-Var Coord (4B)	100.0%	5.3%	0.0%	0.0%

D. Reactive Power Distribution – Scenario 4 [RPi]

--- D1. Volt-Var Local (4A) – per-DER Q statistics ---

98 DERs, 35136 timesteps with Q data

DER	mean Q	std Q	min Q	max Q	% at limit (>0.95*max)
55	-0.0736	0.0187	-0.0784	0.0000	94.0%
56	-0.0736	0.0187	-0.0784	0.0000	94.0%
57	-0.0998	0.0253	-0.1062	0.0000	94.0%
58	-0.2001	0.0508	-0.2130	0.0000	94.0%
59	-0.0998	0.0253	-0.1062	0.0000	94.0%
60	-0.0736	0.0187	-0.0784	0.0000	94.0%
61	-0.0736	0.0187	-0.0784	0.0000	94.0%
62	-0.0998	0.0253	-0.1062	0.0000	94.0%
63	-0.2001	0.0508	-0.2130	0.0000	94.0%
54	-0.1993	0.0507	-0.2130	0.0000	90.1%

Overall Q saturation: mean q_saturated_count = 15.4 of 98 DERs per timestep
Total reactive energy (from summary): 83779.43198871419 MVar.h

--- D2. Volt-Var Local (4A) – Q budget utilisation over time ---

Total |Q| per timestep: mean=9.538 MVar std=2.933 max=14.531 min=0.000
Monthly mean total |Q| (MVar):

Month 1: 7.224
Month 2: 7.962
Month 3: 8.994
Month 4: 10.259
Month 5: 10.727
Month 6: 10.076
Month 7: 10.589
Month 8: 10.489
Month 9: 10.157
Month 10: 9.983
Month 11: 8.031
Month 12: 9.872

--- D1. Volt-Var Coord (4B) – per-DER Q statistics ---

98 DERs, 35136 timesteps with Q data

DER	mean Q	std Q	min Q	max Q	% at limit (>0.95*max)
55	-0.0736	0.0187	-0.0784	0.0000	94.0%
56	-0.0736	0.0187	-0.0784	0.0000	94.0%
58	-0.2001	0.0508	-0.2130	0.0000	94.0%
59	-0.0901	0.0498	-0.1062	0.1061	94.0%
60	-0.0736	0.0187	-0.0784	0.0000	94.0%
61	-0.0638	0.0411	-0.0784	0.0783	93.9%

63	-0.2000	0.0512	-0.2130	0.1897	93.9%
57	-0.0998	0.0253	-0.1062	0.0000	93.9%
62	-0.0996	0.0259	-0.1062	0.1061	93.8%
54	-0.1993	0.0507	-0.2130	0.0000	90.1%

Overall Q saturation: mean q_saturated_count = 17.0 of 98 DERs per timestep
Total reactive energy (from summary): 85936.8700417516 MVar.h

--- D2. Volt-Var Coord (4B) - Q budget utilisation over time ---

Total |Q| per timestep: mean=9.783 MVar std=3.322 max=19.691 min=0.000

Monthly mean total |Q| (MVar):

Month 1: 7.306
Month 2: 8.034
Month 3: 9.235
Month 4: 10.677
Month 5: 11.162
Month 6: 10.187
Month 7: 11.003
Month 8: 10.793
Month 9: 10.586
Month 10: 10.211
Month 11: 8.095
Month 12: 10.010

=====

E. Active Power Curtailment - Scenario 4 [RPI]

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--- E1. Volt-Var Local (4A) - curtailment overview ---

Curtailment steps: 16729

Curtailed energy: 36572.3325 MWh

Curtailment as % of total DER generation: 43.189%

Per-DER curtailment breakdown (top 10 most curtailed):

DER	natural P (MW)	mean P on curt steps	curtailed MWh	curt steps
93	0.4939	1.5140	36.0932	567
91	0.4032	1.2360	29.4638	567
97	0.2016	0.6180	14.7319	567
0	0.1433	0.5475	9.2738	488
96	0.1289	0.4928	8.3464	488
92	0.1218	0.4654	7.8827	488
1	0.0000	0.0280	0.0000	0
2	0.0000	0.0280	0.0000	0
3	0.0000	0.0280	0.0000	0
4	0.0000	0.0280	0.0000	0

Sum of per-DER curtailed energy (curtailment steps only): 105.7919 MWh

Summary curtailed_energy_mwh: 36572.3325 MWh (includes PT1-dynamics P lag on all steps, not just curtailment steps -

Curtailment steps by month:

Month 1: 1113 steps
Month 2: 1035 steps
Month 3: 1332 steps
Month 4: 1290 steps
Month 5: 1734 steps
Month 6: 1044 steps
Month 7: 1576 steps
Month 8: 1335 steps
Month 9: 1352 steps
Month 10: 1586 steps
Month 11: 849 steps
Month 12: 2483 steps

--- E1. Volt-Var Coord (4B) - curtailment overview ---

Curtailment steps: 16729

Curtailed energy: 36556.2071 MWh

Curtailment as % of total DER generation: 43.170%

Per-DER curtailment breakdown (top 10 most curtailed):

DER	natural P (MW)	mean P on curt steps	curtailed MWh	curt steps
93	0.4939	1.5137	36.3512	577
91	0.4032	1.2357	29.6744	577
97	0.2016	0.6178	14.8372	577
0	0.1433	0.5474	9.3045	485
96	0.1289	0.4927	8.3741	485
92	0.1218	0.4653	7.9089	485
1	0.0000	0.0281	0.0000	0
2	0.0000	0.0281	0.0000	0
3	0.0000	0.0281	0.0000	0
4	0.0000	0.0281	0.0000	0

Sum of per-DER curtailed energy (curtailment steps only): 106.4503 MWh

Summary curtailed_energy_mwh: 36556.2071 MWh (includes PT1-dynamics P lag on all steps, not just curtailment steps -

Curtailment steps by month:

Month 1: 1113 steps
Month 2: 1035 steps
Month 3: 1332 steps
Month 4: 1290 steps
Month 5: 1734 steps
Month 6: 1044 steps
Month 7: 1576 steps
Month 8: 1335 steps
Month 9: 1352 steps
Month 10: 1586 steps
Month 11: 849 steps
Month 12: 2483 steps

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F. Export / Import Balance [RPI]

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Note: grid_import_mw < 0 means the network is NET EXPORTING (DER generation exceeds local load + losses).

Scenario	Export steps	Import steps	Export %	Net export MWh	Net import MWh
Baseline	35136	0	100.0%	180828.35	0.00
OLTC	35136	0	100.0%	180708.59	0.00
SVC	35136	0	100.0%	167842.84	0.00
Volt-Var Local (4A)	35136	0	100.0%	144293.62	0.00
Volt-Var Coord (4B)	35136	0	100.0%	144282.47	0.00

Export change vs baseline:

OLTC : -119.77 MWh (-0.1% vs baseline)
SVC : -12985.51 MWh (-7.2% vs baseline)
Volt-Var Local (4A) : -36534.74 MWh (-20.2% vs baseline)
Volt-Var Coord (4B) : -36545.88 MWh (-20.2% vs baseline)

--- F2. Monthly net export comparison: baseline vs Scenario 4 ---

Month	Baseline	Volt-Var Local (4A)	Volt-Var Coord (4B)
1	12737.66	10741.84	10737.83
2	12523.91	10594.29	10588.94
3	14821.42	12028.28	12023.66
4	15642.09	12363.71	12360.18
5	16880.02	12961.56	12973.59
6	14193.53	12464.57	12457.91
7	16581.66	12979.44	12987.87
8	16016.64	13041.43	13041.90
9	15497.71	12024.99	12020.65
10	15508.14	12120.45	12120.75
11	12610.12	10908.60	10907.54
12	17815.46	12064.46	12061.66

(values in MWh exported per month)

=====

G. Violation Frequency Map [RPI]

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--- G. Baseline ---

Total overvoltage bus-steps: 576110 | Total undervoltage bus-steps: 0

Top 10 most frequently overvoltage buses:

Bus	Count	% of all OV steps
98	33011	5.7%
68	32987	5.7%
67	32970	5.7%
66	32952	5.7%
65	32923	5.7%
64	32735	5.7%
63	32583	5.7%
62	32542	5.6%
61	32456	5.6%
60	32195	5.6%

No undervoltage bus-steps recorded.

--- G. OLTC ---

Total overvoltage bus-steps: 245496 | Total undervoltage bus-steps: 0

Top 10 most frequently overvoltage buses:

Bus	Count	% of all OV steps
98	20199	8.2%
68	20186	8.2%
67	20139	8.2%
66	20073	8.2%
65	19961	8.1%
64	19256	7.8%
63	18477	7.5%

62	18261	7.4%
61	17884	7.3%
60	16819	6.9%

No undervoltage bus-steps recorded.

--- G. SVC ---
 Total overvoltage bus-steps: 0 | Total undervoltage bus-steps: 515
 No overvoltage bus-steps recorded.
 Top 10 most frequently undervoltage buses:

Bus	Count	% of all UV steps
98	160	31.1%
65	74	14.4%
66	74	14.4%
67	72	14.0%
68	72	14.0%
64	39	7.6%
63	14	2.7%
62	9	1.7%
61	1	0.2%

--- G. Volt-Var Local (4A) ---
 Total overvoltage bus-steps: 0 | Total undervoltage bus-steps: 0
 No overvoltage bus-steps recorded.
 No undervoltage bus-steps recorded.

--- G. Volt-Var Coord (4B) ---
 Total overvoltage bus-steps: 0 | Total undervoltage bus-steps: 0
 No overvoltage bus-steps recorded.
 No undervoltage bus-steps recorded.

H. Summary Comparison Table [RPI]

Metric	Baseline	OLTC	SVC	Volt-Var Local (4A)	Volt-Var Coord (4B)
Violation steps	33011	20480	35136		
OV bus-steps	576110	245496	0		
UV bus-steps	0	0	515		
VDI	141908.8183	90301.6694	34563.8764		4361
Total losses (MWh)	4672.4490	4792.2141	17657.9589		463
Reactive energy (MVar·h)	—	—	87839.1192		8377
Curtailed energy (MWh)	—	—	—		3657
DER generation (MWh)	84679.7873	84679.7873	84679.7873		8467
Grid export (MWh)	180828.3538	180708.5887	167842.8439		14429
Peak max_vm_pu	1.1166	1.1022	1.0567		
Worst min_vm_pu	1.0086	0.9931	0.9305		
Peak line loading (%)	159.7387	161.9504	189.7993		8
Coordination rate	—	—	—		
Q saturation rate	—	—	—		
Elapsed (s)	1233.1444	1228.0015	2354.5817		1122

--- H2. Loss penalty and reactive cost vs baseline ---
 OLTC : losses +2.6% vs baseline
 SVC : losses +277.9% vs baseline, reactive energy = 87839.12 MVar·h
 Volt-Var Local (4A) : losses -0.8% vs baseline, reactive energy = 83779.43 MVar·h
 Volt-Var Coord (4B) : losses -0.2% vs baseline, reactive energy = 85936.87 MVar·h

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C. Extreme State Timeseries [Laptop]

--- C1. max_vm_pu distribution across scenarios ---

Scenario	mean	median	p95	p99	max	> 1.05 (%)	> 1.08 (%)
Baseline	1.0769	1.0766	1.0962	1.1038	1.1166	99.3%	37.1%
OLTC	1.0622	1.0619	1.0816	1.0893	1.1022	86.5%	6.7%
SVC	1.0278	1.0250	1.0398	1.0486	1.0567	0.6%	0.0%
Volt-Var Local (4A)	1.0357	1.0354	1.0515	1.0591	1.0600	5.3%	0.0%
Volt-Var Coord (4B)	1.0354	1.0352	1.0515	1.0591	1.0600	5.3%	0.0%

--- C2. max_line_loading_pct distribution across scenarios ---

Scenario	mean	median	p95	p99	max	> 100% (steps)
Baseline	78.82	73.15	127.20	141.17	159.74	7515
OLTC	79.93	74.19	128.99	143.14	161.95	7919
SVC	131.31	126.72	161.37	173.82	189.80	35136
Volt-Var Local (4A)	79.78	85.41	89.62	89.92	90.00	0
Volt-Var Coord (4B)	79.77	85.39	89.62	89.92	90.00	0

--- C3. Voltage envelope: how often does max_vm_pu exceed thresholds? ---

Scenario	>= 1.02	>= 1.05	>= 1.08	>= 1.10
Baseline	100.0%	99.3%	37.1%	2.2%
OLTC	100.0%	86.5%	6.7%	0.0%
SVC	100.0%	0.6%	0.0%	0.0%
Volt-Var Local (4A)	100.0%	5.3%	0.0%	0.0%
Volt-Var Coord (4B)	100.0%	5.3%	0.0%	0.0%

D. Reactive Power Distribution – Scenario 4 [Laptop]

--- D1. Volt-Var Local (4A) – per-DER Q statistics ---

98 DERs, 35136 timesteps with Q data

DER	mean Q	std Q	min Q	max Q	% at limit (>0.95*max)
55	-0.0736	0.0187	-0.0784	0.0000	94.0%
56	-0.0736	0.0187	-0.0784	0.0000	94.0%
57	-0.0998	0.0253	-0.1062	0.0000	94.0%
58	-0.2001	0.0508	-0.2130	0.0000	94.0%
59	-0.0998	0.0253	-0.1062	0.0000	94.0%
60	-0.0736	0.0187	-0.0784	0.0000	94.0%
61	-0.0736	0.0187	-0.0784	0.0000	94.0%
62	-0.0998	0.0253	-0.1062	0.0000	94.0%
63	-0.2001	0.0508	-0.2130	0.0000	94.0%
54	-0.1992	0.0507	-0.2130	0.0000	90.0%

Overall Q saturation: mean q_saturated_count = 15.4 of 98 DERs per timestep
Total reactive energy (from summary): 83751.37597911102 MVar.h

--- D2. Volt-Var Local (4A) – Q budget utilisation over time ---

Total |Q| per timestep: mean=9.535 MVar std=2.932 max=14.526 min=0.000

Monthly mean total |Q| (MVar):

Month 1: 7.222
Month 2: 7.959
Month 3: 8.991
Month 4: 10.255
Month 5: 10.724
Month 6: 10.073
Month 7: 10.586
Month 8: 10.485
Month 9: 10.154
Month 10: 9.979
Month 11: 8.028
Month 12: 9.869

--- D1. Volt-Var Coord (4B) – per-DER Q statistics ---

98 DERs, 35136 timesteps with Q data

DER	mean Q	std Q	min Q	max Q	% at limit (>0.95*max)
55	-0.0736	0.0187	-0.0784	0.0000	94.0%
56	-0.0736	0.0187	-0.0784	0.0000	94.0%
58	-0.2001	0.0508	-0.2130	0.0000	94.0%
59	-0.0900	0.0500	-0.1062	0.1062	94.0%
60	-0.0736	0.0187	-0.0784	0.0000	94.0%
63	-0.2001	0.0510	-0.2130	0.2130	93.9%
61	-0.0638	0.0412	-0.0784	0.0784	93.9%
57	-0.0998	0.0253	-0.1062	0.0000	93.9%
62	-0.0996	0.0260	-0.1062	0.1062	93.8%
54	-0.1992	0.0507	-0.2130	0.0000	90.0%

Overall Q saturation: mean q_saturated_count = 17.1 of 98 DERs per timestep
Total reactive energy (from summary): 85932.71566368979 MVar.h

--- D2. Volt-Var Coord (4B) – Q budget utilisation over time ---

Total |Q| per timestep: mean=9.783 MVar std=3.326 max=19.695 min=0.000

Monthly mean total |Q| (MVar):

Month 1: 7.306
Month 2: 8.034
Month 3: 9.232
Month 4: 10.679
Month 5: 11.161
Month 6: 10.186
Month 7: 11.003
Month 8: 10.791
Month 9: 10.589
Month 10: 10.210
Month 11: 8.093
Month 12: 10.010

E. Active Power Curtailment – Scenario 4 [Laptop]

--- E1. Volt-Var Local (4A) – curtailment overview ---

Curtailment steps: 16727
 Curtailed energy: 36568.4386 MWh
 Curtailment as % of total DER generation: 43.184%

Per-DER curtailment breakdown (top 10 most curtailed):

DER	natural P (MW)	mean P on curt steps	curtailed MWh	curt steps
93	0.4944	1.5142	36.1405	567
91	0.4036	1.2360	29.5025	567
97	0.2018	0.6180	14.7512	567
0	0.1433	0.5476	9.2738	488
96	0.1289	0.4929	8.3464	488
92	0.1218	0.4655	7.8827	488
1	0.0000	0.0280	0.0000	0
2	0.0000	0.0280	0.0000	0
3	0.0000	0.0280	0.0000	0
4	0.0000	0.0280	0.0000	0

Sum of per-DER curtailed energy (curtailment steps only): 105.8972 MWh

Summary curtailed_energy_mwh: 36568.4386 MWh (includes PT1-dynamics P lag on all steps, not just curtailment steps -

Curtailment steps by month:

Month 1: 1113 steps
 Month 2: 1034 steps
 Month 3: 1332 steps
 Month 4: 1291 steps
 Month 5: 1734 steps
 Month 6: 1044 steps
 Month 7: 1576 steps
 Month 8: 1333 steps
 Month 9: 1352 steps
 Month 10: 1586 steps
 Month 11: 849 steps
 Month 12: 2483 steps

--- E1. Volt-Var Coord (4B) -- curtailment overview ---

Curtailment steps: 16727

Curtailed energy: 36552.7073 MWh

Curtailment as % of total DER generation: 43.166%

Per-DER curtailment breakdown (top 10 most curtailed):

DER	natural P (MW)	mean P on curt steps	curtailed MWh	curt steps
93	0.4944	1.5136	36.3596	577
91	0.4036	1.2356	29.6813	577
97	0.2018	0.6178	14.8406	577
0	0.1433	0.5474	9.3003	485
96	0.1289	0.4926	8.3703	485
92	0.1218	0.4653	7.9053	485
1	0.0000	0.0281	0.0000	0
2	0.0000	0.0281	0.0000	0
3	0.0000	0.0281	0.0000	0
4	0.0000	0.0281	0.0000	0

Sum of per-DER curtailed energy (curtailment steps only): 106.4574 MWh

Summary curtailed_energy_mwh: 36552.7073 MWh (includes PT1-dynamics P lag on all steps, not just curtailment steps -

Curtailment steps by month:

Month 1: 1113 steps
 Month 2: 1034 steps
 Month 3: 1332 steps
 Month 4: 1291 steps
 Month 5: 1734 steps
 Month 6: 1044 steps
 Month 7: 1576 steps
 Month 8: 1333 steps
 Month 9: 1352 steps
 Month 10: 1586 steps
 Month 11: 849 steps
 Month 12: 2483 steps

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F. Export / Import Balance [Laptop]

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Note: grid_import_mw < 0 means the network is NET EXPORTING (DER generation exceeds local load + losses).

Scenario	Export steps	Import steps	Export %	Net export MWh	Net import MWh
Baseline	35136	0	100.0%	180828.35	0.00
OLTC	35136	0	100.0%	180708.59	0.00
SVC	35136	0	100.0%	167842.84	0.00
Volt-Var Local (4A)	35136	0	100.0%	144298.05	0.00
Volt-Var Coord (4B)	35136	0	100.0%	144286.13	0.00

Export change vs baseline:

OLTC	: -119.77 MWh (-0.1% vs baseline)
SVC	: -12985.51 MWh (-7.2% vs baseline)
Volt-Var Local (4A)	: -36530.30 MWh (-20.2% vs baseline)
Volt-Var Coord (4B)	: -36542.22 MWh (-20.2% vs baseline)

--- F2. Monthly net export comparison: baseline vs Scenario 4 ---

Month	Baseline	Volt-Var Local (4A)	Volt-Var Coord (4B)
1	12737.66	10742.94	10738.00
2	12523.91	10594.94	10589.14
3	14821.42	12028.94	12024.31
4	15642.09	12364.41	12359.51
5	16880.02	12961.67	12974.59
6	14193.53	12464.57	12457.47
7	16581.66	12979.76	12989.09
8	16016.64	13041.73	13043.47
9	15497.71	12025.90	12021.22
10	15508.14	12121.16	12120.73
11	12610.12	10908.17	10907.58
12	17815.46	12063.86	12061.02

(values in MWh exported per month)

G. Violation Frequency Map [Laptop]

--- G. Baseline ---

Total overvoltage bus-steps: 576110 | Total undervoltage bus-steps: 0

Top 10 most frequently overvoltage buses:

Bus	Count	% of all OV steps
98	33011	5.7%
68	32987	5.7%
67	32970	5.7%
66	32952	5.7%
65	32923	5.7%
64	32735	5.7%
63	32583	5.7%
62	32542	5.6%
61	32456	5.6%
60	32195	5.6%

No undervoltage bus-steps recorded.

--- G. OLTC ---

Total overvoltage bus-steps: 245496 | Total undervoltage bus-steps: 0

Top 10 most frequently overvoltage buses:

Bus	Count	% of all OV steps
98	20199	8.2%
68	20186	8.2%
67	20139	8.2%
66	20073	8.2%
65	19961	8.1%
64	19256	7.8%
63	18477	7.5%
62	18261	7.4%
61	17884	7.3%
60	16819	6.9%

No undervoltage bus-steps recorded.

--- G. SVC ---

Total overvoltage bus-steps: 0 | Total undervoltage bus-steps: 515

No overvoltage bus-steps recorded.

Top 10 most frequently undervoltage buses:

Bus	Count	% of all UV steps
98	160	31.1%
65	74	14.4%
66	74	14.4%
67	72	14.0%
68	72	14.0%
64	39	7.6%
63	14	2.7%
62	9	1.7%
61	1	0.2%

--- G. Volt-Var Local (4A) ---

Total overvoltage bus-steps: 0 | Total undervoltage bus-steps: 0

No overvoltage bus-steps recorded.

No undervoltage bus-steps recorded.

--- G. Volt-Var Coord (4B) ---

Total overvoltage bus-steps: 0 | Total undervoltage bus-steps: 0

No overvoltage bus-steps recorded.

No undervoltage bus-steps recorded.

=====

H. Summary Comparison Table [Laptop]

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Metric	Baseline	OLTC	SVC	Volt-Var	Local
Violation steps	33011	20480	35136		
OV bus-steps	576110	245496	0		
UV bus-steps	0	0	515		
VDI	141908.8183	90301.6694	34563.8764		4363
Total losses (MWh)	4672.4490	4792.2141	17657.9589		463
Reactive energy (MVar·h)	—	—	87839.1192		8375
Curtailed energy (MWh)	—	—	—		3656
DER generation (MWh)	84679.7873	84679.7873	84679.7873		8467
Grid export (MWh)	180828.3538	180708.5887	167842.8439		14429
Peak max_vm_pu	1.1166	1.1022	1.0567		
Worst min_vm_pu	1.0086	0.9931	0.9305		
Peak line loading (%)	159.7387	161.9504	189.7993		8
Coordination rate	—	—	—		
Q saturation rate	—	—	—		
Elapsed (s)	1989.0767	1970.2400	3672.6164		923

--- H2. Loss penalty and reactive cost vs baseline ---

OLTC : losses +2.6% vs baseline
SVC : losses +277.9% vs baseline, reactive energy = 87839.12 MVar·h
Volt-Var Local (4A) : losses -0.8% vs baseline, reactive energy = 83751.38 MVar·h
Volt-Var Coord (4B) : losses -0.2% vs baseline, reactive energy = 85932.72 MVar·h

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END OF REPORT

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Q_initial clip diagnostic (H1/H2)

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Q_INITIAL CLIP DIAGNOSTIC - scenario: volt_var_coord
=====
```

```
RPi records:      35136
Laptop records: 35136
Common timesteps: 35136
```

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PART A - q_saturated_count (out of up to 98 DERs), RPi vs Laptop
-----
```

```
RPi      : mean=17.01  median=18.0  max=86  min=0
Laptop   : mean=17.11  median=18.0  max=87  min=0
```

```
Per-timestep comparison (n=35136):
  RPi higher      : 1153 (3.3%)
  Laptop higher   : 2194 (6.2%)
  Equal           : 31789 (90.5%)
  Mean diff (RPi - Laptop): -0.097 DERs
```

```
>> Saturation counts are in the same order of magnitude.
>> Consistent with H1 (truncation pushing values over the
clip boundary only marginally more often on RPi).
```

```
-----
PART B - per-DER |q_mvar| magnitude comparison (q_mvar_by_sgen)
-----
```

```
Sampling 2067 of 35136 common timesteps (every 17th)
```

```
|q_rpi - q_laptop| across 202566 (DER, t) pairs:
  mean=0.000143  median=0.000063  max=0.402737 (MVar)
```

```
|q_rpi| / |q_laptop| ratio (excluding near-zero laptop values):
  mean=1.0008  median=1.0000
  fraction of ratios within +-10% of 1.0: 99.9%
```

```
>> The vast majority of (DER, t) pairs agree to within +-10%.
>> Consistent with H1 (truncation only bites at the margin
for a small subset of DERs/timesteps).
```

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PART C - top 15 largest individual |q_rpi - q_laptop| disagreements
-----
```

t= 32759	DER= 63	q_rpi=+0.18971	q_laptop=-0.21303	diff=0.40274	[sat_count RPi=72 Laptop=76]
t= 32657	DER= 63	q_rpi=+0.12073	q_laptop=-0.21303	diff=0.33376	[sat_count RPi=81 Laptop=84]
t= 13957	DER= 62	q_rpi=+0.10608	q_laptop=-0.10624	diff=0.21233	[sat_count RPi=73 Laptop=77]
t= 1547	DER= 62	q_rpi=+0.10608	q_laptop=-0.10624	diff=0.21233	[sat_count RPi=81 Laptop=87]
t= 32759	DER= 61	q_rpi=-0.07835	q_laptop=+0.07835	diff=0.15670	[sat_count RPi=72 Laptop=76]
t= 32657	DER= 61	q_rpi=-0.07835	q_laptop=+0.07835	diff=0.15670	[sat_count RPi=81 Laptop=84]
t= 18972	DER= 63	q_rpi=-0.21303	q_laptop=-0.08224	diff=0.13080	[sat_count RPi=66 Laptop=76]
t= 32657	DER= 62	q_rpi=-0.01839	q_laptop=+0.10624	diff=0.12463	[sat_count RPi=81 Laptop=84]
t= 32759	DER= 62	q_rpi=+0.02541	q_laptop=+0.10624	diff=0.08083	[sat_count RPi=72 Laptop=76]
t= 20502	DER= 93	q_rpi=-1.86401	q_laptop=-1.82255	diff=0.04146	[sat_count RPi=44 Laptop=40]
t= 20502	DER= 43	q_rpi=-0.14385	q_laptop=-0.11037	diff=0.03347	[sat_count RPi=44 Laptop=40]
t= 20502	DER= 91	q_rpi=-1.52060	q_laptop=-1.49067	diff=0.02993	[sat_count RPi=44 Laptop=40]
t= 20502	DER= 0	q_rpi=-0.08279	q_laptop=-0.05388	diff=0.02891	[sat_count RPi=44 Laptop=40]
t= 23749	DER= 93	q_rpi=-1.63327	q_laptop=-1.60465	diff=0.02861	[sat_count RPi=82 Laptop=75]
t= 13957	DER= 61	q_rpi=+0.04983	q_laptop=+0.07835	diff=0.02852	[sat_count RPi=73 Laptop=77]

```
=====
END OF REPORT
=====
```

Coordination ground-truth check

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=====
COORDINATION GROUND-TRUTH CHECK - scenario: volt_var_coord
Thresholds: V_BAND=[0.95, 1.05] MIN_VIOLATION_DV=0.001
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RPI - ground-truth check on coordination_active timesteps
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Timesteps with coordination_active=True: 33008 of 35136

Of 33008 active timesteps with usable vm_pu_by_bus data:
  Real violation remained (margin < 0) : 0 (0.0%)
  Near the trigger's own noise floor (0 <= margin < 0.001) : 0 (0.0%)
  Comfortably inside the band (margin >= 0.001) : 33008 (100.0%)

Margin distribution (pu, worst bus per timestep):
  mean=0.01585 median=0.01512 min=0.00216 max=0.02500

>> No real post-coordination violations found on any active timestep.
>> The coordinator's corrections (even when triggered by a
    margin-sensitive linear prediction) are landing in a safe
    final state every time it activates.

-----

Laptop - ground-truth check on coordination_active timesteps
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Timesteps with coordination_active=True: 2277 of 35136

Of 2277 active timesteps with usable vm_pu_by_bus data:
  Real violation remained (margin < 0) : 0 (0.0%)
  Near the trigger's own noise floor (0 <= margin < 0.001) : 0 (0.0%)
  Comfortably inside the band (margin >= 0.001) : 2277 (100.0%)

Margin distribution (pu, worst bus per timestep):
  mean=0.02240 median=0.02500 min=0.00599 max=0.02500

>> No real post-coordination violations found on any active timestep.
>> The coordinator's corrections (even when triggered by a
    margin-sensitive linear prediction) are landing in a safe
    final state every time it activates.

=====
END OF REPORT
=====
```